

Staged predictive modelling of PEGDA free-radical photopolymerization

Matthias Wessling

Master's-thesis-style final report

AVT.CVT — Photopolymerization modelling campaign

RWTH Aachen University

13 September 2026

Abstract

This report is a master’s-thesis-style account of a staged campaign to predict acrylate conversion, operational gelation, and loop-aware hydrogel modulus for poly(ethylene glycol) diacrylate (PEGDA) photopolymerization without treating those quantities as a single dose number. The locked scientific object is chain-growth free-radical polymerization driven by a prescribed irradiance history, later a digital micromirror surface field and Beer–Lambert attenuation, later plug flow, and finally a Macosko–Miller network post-processor with intramolecular loops. Production Type I kinetics follow Montgomery, Hamel, Skovran and Qi (*Extreme Mechanics Letters*, 2022) for PEGDA 250 / Irgacure 819. Projection uses their Gaussian pixel kernel as a fallback, not as a measured point-spread function. Plug flow uses a *separate* constitutive book taken from Slutzky, Stone and Nunes (*Soft Matter*, 2019) for a PEGDA 575 jet. Network mechanics follow Zhu, Yang, Genin, Lu, Xu and Lin (*Journal of the Mechanics and Physics of Solids*, 2020), with eosin Y bleaching ordinary differential equations switched off. The high-fidelity species model of Dobson and Bowman (*Advanced Functional Materials*, 2024) is a reference ontology, not the everyday right-hand side.

Seven stages (0–6) were implemented in a Python package under `src/photopolymerization/`, each with a frozen configuration, forward run, named validator, tests, overview figure, and journal tag. As of 13 September 2026 all seven stages are internally **validated**. None is **matched** to Fourier-transform infrared conversion, a camera kernel, an experimental critical speed, or rheology. Dissolved oxygen in the Montgomery book remains **still required**. Internal findings that survive those caveats include failed equal-dose reciprocity in the Montgomery closures, recovery of Slutzky’s oxygen–monomer identity during induction, a grayscale interface width that is dominated by the optical kernel rather than Table 3 radical diffusion on an eight-second strip, numerical agreement of a Fig. 3-style jet with $M/M_0 \simeq 0.86$, and a loop-reduced elastically active fraction $\eta = 0.378$ at full conversion for 20 vol% PEGDA with Zhu’s fitted loop parameter.

The report is not a printer manual and not a claim that in-house AVT resin equals any of the three literature formulations. The recommended next scientific step is a Stage 1 FTIR overlay on a named recipe with a pre-declared error band.

Contents

1	Introduction	6
1.1	Motivation	6
1.2	Locked scientific object and non-goals	7
1.3	Intended reader	7
1.4	How this report is organised	7
1.5	Statement of originality and limitations	7
2	Chemical and process background	8
2.1	What we are trying to predict	8
2.2	The chemical inventory	9
2.2.1	Photoinitiators: Type I versus Type II	9
2.2.2	Monomer, conversion, and what FTIR actually measures	10
2.2.3	Radicals as a short-lived catalyst, not a stored dose	10
2.2.4	Oxygen as a reagent, not a correction factor	11
2.3	The kinetic skeleton (chain growth)	11
2.3.1	Why k_p and k_t are not constants	11
2.3.2	Thiol–ene is a different polymerization	12
2.4	Dose is not a state: the failure of exposure reciprocity	12
2.5	Light as a field in a participating medium	13
2.5.1	Beer–Lambert attenuation that evolves with chemistry	13
2.5.2	Pixels, grayscale, and the point-spread function	13
2.6	Two independent motions	14
2.7	Flow replenishes oxygenated resin	15
2.8	Conversion is not gelation is not modulus	15
2.8.1	Gelation	15
2.8.2	Loops and elastically active density	16
2.8.3	Heat, shrinkage, and swelling	17

2.9	How the five campaign papers partition the ontology	17
2.10	What must not be identified	17
2.11	Implications for a predictive description	18
3	Literature study and research gaps	19
3.1	Scope of the corpus	19
3.2	High-fidelity kinetics: Dobson and Bowman	19
3.3	Production spatial kinetics: Montgomery et al.	19
3.4	Projection kinematics: Meenakshisundaram, Sturm and Williams	20
3.5	Plug flow: Slutzky, Stone and Nunes	20
3.6	Network mechanics: Zhu et al.	21
3.7	Supporting literature that the five papers rest on	21
3.8	Working curves, CLIP, and why they are not the polymer state	22
3.9	Undermind synthesis versus what was coded	22
3.10	The gap closed by this campaign	23
4	Software architecture and numerical methods	24
4.1	Design principles	24
4.2	Repository layout	24
4.3	The campaign quartet	25
4.4	Parameter books	26
4.4.1	Montgomery 2022 (Type I, PEGDA 250)	26
4.4.2	Slutzky 2019 (Type I, PEGDA 575 jet)	26
4.4.3	Zhu 2020 (network only)	26
4.5	Time integration of the local ODE	26
4.6	Projection without chemistry	26
4.7	One-dimensional reaction–diffusion	27
4.8	Plug-flow marching	27
4.9	Network algebra	27
4.10	Two Type I right-hand sides that must not be merged	27
4.11	What is deliberately absent	28
5	Results of the staged campaign	29
5.1	Stage 0 — provenance before kinetics	30
5.2	Stage 1 — local Montgomery ODE	30
5.3	Stage 2 — analytic identities on a constant-rate fork	30

5.4	Stage 3 — Gaussian projection	31
5.5	Stage 4 — reaction–diffusion and Beer–Lambert	31
5.6	Stage 5 — Slutzky plug flow	33
5.7	Stage 6 — Zhu network post-processor	34
5.8	Cross-stage reasonability	34
5.9	Tests	35
6	What would constitute a literature match	36
6.1	Recommended order	36
6.2	Stage 1	36
6.3	Stage 3	37
6.4	Stage 4	37
6.5	Stage 5	37
6.6	Stage 6	37
6.7	Forbidden upgrades	37
7	Conclusions, limitations, and parked work	38
7.1	What this campaign established	38
7.2	Limitations	38
7.3	Parked work (do not auto-start)	39
7.4	Immediate scientific next step	39
8	Nomenclature	40
9	Appendix: verification log and file inventory	41
9.1	How to reproduce the internal gates	41
9.2	Overview figures	41
9.3	Journal entries	42
9.4	What is gitignored	42

List of Figures

2.1	Schematic chemical inventory for a free-radical PEGDA photopolymerization. Boxes are species classes, not a complete kinetic scheme. Conversion p counts reacted acrylate groups; it is not yet gelation or modulus.	9
2.2	Schematic only. Two protocols with the same nominal dose need not yield the same conversion when termination is bimolecular and oxygen participates [1–3]. The curves are not experimental measurements.	13
2.3	Left: the illumination pattern moves over a vat (scanning mask or scrolled image) while the resin is, to first approximation, still [4]. Right: the resin flows through a fixed light window [5]. A fluid element experiences the <i>relative</i> motion of pattern and material.	14
2.4	Four layers that must not be identified. Light history and species fields determine conversion p ; p and composition determine whether a sample-spanning network exists and how many chains are elastically active; modulus is a mechanical consequence of that network [6–8].	16
5.1	Locked stage ladder. Parked items (Dobson production RHS, NS–gel, inverse DMD, Type II bleaching, Jacobs as polymer state) remain parked.	29
5.2	Stage 0 provenance overview (teaching layout of the book, not an experiment). .	30
5.3	Stage 1 local ODE: conversion histories, equal-dose pair, and oxygen induction. .	31
5.4	Stage 2: oxygen–monomer identity and quasi-steady radical residual.	32
5.5	Stage 3 Gaussian projection (Montgomery Eq. 7 fallback). No conversion is plotted.	32
5.6	Stage 4: optical front, oxygen wall, and G0 G80 strip width.	33
5.7	Stage 5 plug flow on the Slutsky book: conversion along x , U_c trends, pulses. . .	34
5.8	Stage 6 Macosko–Miller loops: $P_{\text{gel}}(c)$, $\eta(p)$, and a hypothetical Montgomery trajectory.	35

List of Tables

2.1	Division of labour among the five unique reports in the campaign corpus. “Owns” means: this is the layer the paper actually closes with equations and experiments. It does not mean the other layers are false.	17
5.1	Campaign status on 13 September 2026. Internal validated is not a literature overlay.	29
8.1	Principal symbols.	40

Chapter 1

Introduction

1.1 Motivation

This report is a master’s-thesis-style account of a staged modelling campaign for *free-radical photopolymerization of poly(ethylene glycol) diacrylate* (PEGDA). The engineering context is continuous-flow digital lithography of multi-porosity hydrogel patches, as envisaged in the DFG proposal *Continuous-Flow Digital Lithography of Multi-Porosity Hydrogel Patches* (AVT.CVT, RWTH Aachen). The scientific problem is older and more elementary than that device: given a known light history, and later a known flow, can one predict how many acrylate groups have reacted, whether a sample-spanning network exists, and how stiff the gel is—without treating those three questions as the same number?

A machine operator can print without answering them. A working curve relating cure depth to the logarithm of dose [9] is a calibration for a given resin, lamp, and speed. It is not a constitutive law. Two parcels of resin can receive the same

$$E = \int I(t) dt \tag{1.1}$$

and still differ in conversion p , gelation, and modulus, because bimolecular termination and oxygen inhibition make the outcome depend on the *shape* of $I(t)$ [1–3]. A flowing jet continuously replenishes oxygenated liquid [5]. A DMD pixel is not a top-hat of light [4, 10]. A dilute PEGDA solution may never gel even at full acrylate conversion, because too many additions close loops [6].

The campaign therefore refuses a single mixed three-dimensional code on day one. It implements seven stages (0–6), each with a frozen configuration, a forward run, named pass/fail gates, and one evidence tag (**proposed**, **smoke-tested**, **validated**, **matched**, or **invalid**). As of 13 September 2026, Stages 0–6 are **validated** internally. None is **matched** to FTIR, a camera point-spread function, an experimental critical speed, or rheology.

1.2 Locked scientific object and non-goals

Object. PEGDA *chain-growth* free-radical photopolymerization: prescribed irradiance $I(t)$ maps to conversion p ; later a DMD surface field and Beer–Lambert attenuation; later plug flow; then a loop-aware network post-processor $\eta(p, [\text{PEGDA}]_0)$ and phantom modulus G .

Non-goals (parked). The full multi-species free-volume model of Dobson and Bowman [3] as an everyday right-hand side; Navier–Stokes coupled to gelation; scattering optics; inverse DMD design; PNIPAM and thiol–ene closures; mixing eosin Y Type II bleaching ODEs into an Irgacure Type I rate law; using a Jacobs working curve as the polymer state.

Two velocities. Pattern velocity $\mathbf{v}_{\text{pattern}}$ (scan or mask) is independent of resin velocity $\mathbf{u}$. A fluid element sees $\mathbf{v}_{\text{pattern}} - \mathbf{u}$ [4, 5].

1.3 Intended reader

The text is written for a starting graduate student who knows physical chemistry and transport but need not have worked on vat photopolymerization. Jargon is defined once. Evidence tags are used in the same sense throughout: engineering success (the solver finished) is not scientific validity.

1.4 How this report is organised

Chapter 2 is the ontology: species, rates, light, oxygen, flow, and network, and the list of quantities that must not be identified. Chapter 3 is a literature study of the five campaign papers and the supporting oxygen, reciprocity, and gelation literature, including the gap that the code was written to close. Chapter 4 describes the Python package, the campaign quartet, numerical methods, and parameter books. Chapter 5 reports Stages 0–6 with gates, figures, and what each stage is *not*. Chapter 6 states what would be required to retag any stage **matched**. Chapter 7 summarises findings, limitations, and parked work.

1.5 Statement of originality and limitations

No new laboratory experiment is reported. The contribution is a citable modelling stack with separate chemistry keys, internal gates, and an explicit refusal to mix recipes. Where a paper figure is reproduced only qualitatively (Montgomery Fig. 3-style Gaussians; Slutzky Figs. 2–3 illustrations), the tag remains **validated**, not **matched**. Publisher PDFs in **Literature/** are local-only and are not redistributed.

Chapter 2

Chemical and process background

This chapter is the chemistry and process ontology of the campaign. It is written for a reader with undergraduate physical chemistry and transport who has not necessarily worked on stereolithography. Quantitative claims about a specific PEGDA formulation remain **still required** until they are taken from a named experiment or paper figure under the same recipe. Figures in this chapter are teaching cartoons, not experimental measurements.

2.1 What we are trying to predict

A photopolymerizable liquid is a mixture that stays liquid in the dark and becomes a covalently connected solid (a gel or a densely crosslinked plastic) where and when it is illuminated, provided enough of the right chemical events occur. The engineering goal is to *place* that solid: in a vat printer, as a layer of a part; in a microfluidic channel, as a fiber or a patch; in a hydrogel, as a spatial map of stiffness.

A tempting one-number description is the *dose*

$$E = \int I(t) dt, \tag{2.1}$$

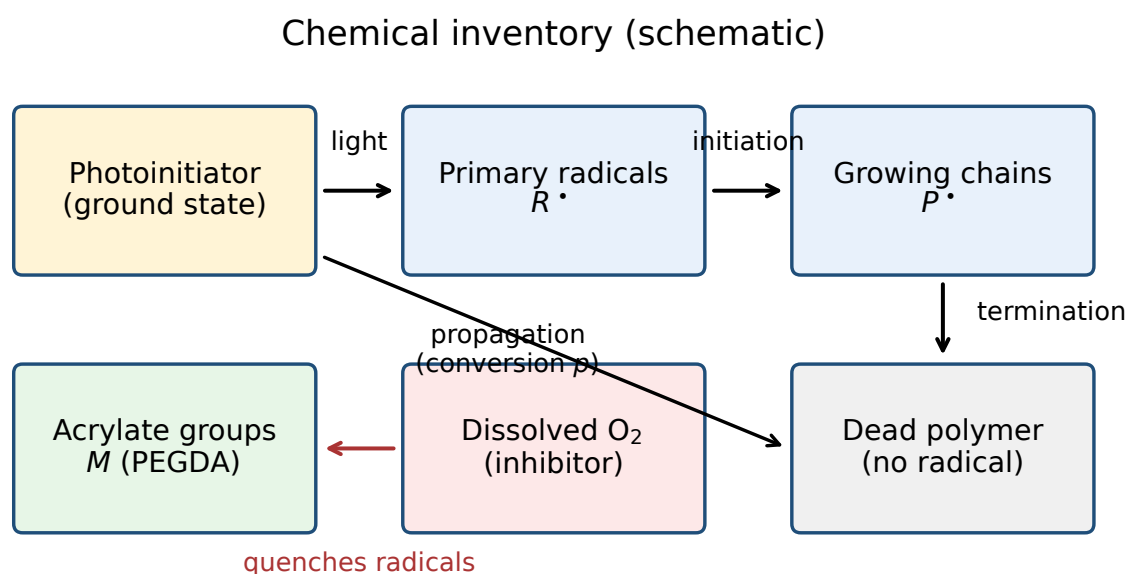
where I is the irradiance (energy, or moles of photons, arriving per area per time) at a point on the resin. Stereolithography practice has long used a threshold form of this idea: a working curve relating cure depth to $\ln(E/E_c)$ [9]. That description is useful as a machine calibration for a *given* resin, optics, and speed. It is not a material state. Two parcels of resin can receive the same E and still differ in how many acrylate groups have reacted, whether a sample-spanning network exists, and how stiff the gel is. The rest of this note explains why, by naming the chemical inventory, the competing rates, the transport of light and oxygen, and the difference between chemical conversion and a mechanically active network.

PEGDA as the running example. Poly(ethylene glycol) diacrylate (PEGDA) is a short chain with an acrylate group at each end. Each acrylate carries a carbon–carbon double bond that can add to a radical. Because there are two such groups per molecule, chain growth can connect different PEGDA molecules into a branched, then crosslinked, network [5, 6, 10]. PEGDA

is *not* one resin: molecular weight (for example ≈ 250 versus $\approx 575 \text{ g mol}^{-1}$), water content, photoinitiator, and atmosphere change both the kinetics and the gel point. Numbers fitted to one formulation must not be treated as universal PEGDA constants.

2.2 The chemical inventory

Figure 2.1 is a cartoon of the species that matter for a typical free-radical acrylate photopolymerization. Light does not “glue PEGDA together” in one step. Light is absorbed by a *photoinitiator*. The initiator produces *radicals*. Radicals add to acrylate double bonds (*propagation*), which is the step that consumes monomer and builds polymer. Radicals also destroy one another (*termination*) and are scavenged by dissolved oxygen (*inhibition*). Only the competition among these processes decides how much conversion occurs for a given illumination.



Light does not polymerize PEGDA directly. It generates radicals that compete for acrylate addition versus oxygen scavenging and radical-radical termination.

Figure 2.1: Schematic chemical inventory for a free-radical PEGDA photopolymerization. Boxes are species classes, not a complete kinetic scheme. Conversion p counts reacted acrylate groups; it is not yet gelation or modulus.

2.2.1 Photoinitiators: Type I versus Type II

Photoinitiators are classified by how they generate the first radicals after absorbing a photon [6, 11].

Type I A single molecule undergoes bond cleavage and yields (typically two) initiating radicals. Irgacure 819 (phenylbis(2,4,6-trimethylbenzoyl)phosphine oxide), used by Montgomery

et al. [10] and in the illustrations of Dobson and Bowman [3], is in this class. A compact description of photolysis is



with a rate proportional to the local irradiance and to the remaining initiator concentration [10].

Type II The absorber is excited to a triplet and must react with a co-initiator (often an amine) to produce radicals. Eosin Y with triethanolamine, the system of Zhu et al. [6], is Type II. The excited dye can also return to the ground state or be irreversibly bleached to a leuco product. Bleaching therefore changes *both* the initiation rate *and* the optical absorption of the sample as curing proceeds.

These are different chemical objects. A rate law written for Irgacure cleavage is not interchangeable with an eosin Y bleaching cascade, even if both are used to cure PEGDA.

2.2.2 Monomer, conversion, and what FTIR actually measures

Let $[M]$ be the molar concentration of unreacted acrylate *functional groups* (not necessarily the concentration of PEGDA molecules). The degree of conversion is

$$p(t) = 1 - \frac{[M](t)}{[M]_0}. \quad (2.3)$$

In practice p is often inferred from the decay of the infrared acrylate band relative to an internal carbonyl standard [10]. That measurement reports *how many double bonds have disappeared*. It does not, by itself, report whether those additions connected two different chains or closed a loop on the same chain, nor whether the sample has gelled.

2.2.3 Radicals as a short-lived catalyst, not a stored dose

The radical concentration $[R^\bullet]$ is typically orders of magnitude smaller than $[M]$ or $[O_2]$. Radicals are generated by initiation and lost by termination and inhibition. If generation and loss nearly balance, one may write a quasi-steady estimate such as

$$[R^\bullet] \approx \sqrt{\frac{R_i}{2k_t}} \quad (2.4)$$

in the oxygen-free, bimolecular-termination limit, with R_i the initiation rate and k_t the termination coefficient [6]. Equation (2.4) is an *approximation*, useful for insight. It fails during the first instants after the lamp is switched on, when oxygen is still abundant, and whenever k_t itself depends strongly on conversion. Keeping the radical balance as a genuine time-dependent equation is therefore a modelling choice with chemical meaning: one refuses to assume that radicals forget their recent history [3, 5, 10].

2.2.4 Oxygen as a reagent, not a correction factor

Molecular oxygen reacts rapidly with carbon-centred radicals to give comparatively unreactive peroxy species [5, 10, 12, 13]:



Until the local oxygen inventory is depleted, propagation cannot compete. This has two engineering consequences.

First, there is an *induction period*: illumination can proceed with almost no conversion while dissolved oxygen is consumed. Second, oxygen is *replenished* by diffusion from an interface (air, an oxygen-permeable window such as PDMS, or a flowing liquid that has not yet been illuminated). Dendukuri et al. [13] showed that in a PDMS microfluidic device the oxygen leaking through the elastomer sets a dead zone and a delay that cannot be read from a well-mixed bottle experiment. Tumbleston et al. [14] turned a persistent oxygen-inhibited liquid layer into a manufacturing feature (continuous liquid interface production). The ontology point is the same: O_2 is a transported chemical field with sources at boundaries, not a constant “inhibition factor” multiplying the dose.

2.3 The kinetic skeleton (chain growth)

A minimal reaction list, following the scheme used in flow by Slutzky et al. [5] and in grayscale printing by Montgomery et al. [10], is:

Photolysis / initiation	$PI \xrightarrow{h\nu} R^\bullet$
Chain initiation	$R^\bullet + M \rightarrow P^\bullet$
Propagation	$P_n^\bullet + M \xrightarrow{k_p} P_{n+1}^\bullet$
Bimolecular termination	$P^\bullet + P^\bullet \xrightarrow{k_t} \text{dead polymer}$
Oxygen inhibition	$P^\bullet + O_2 \xrightarrow{k_O} \text{inactive}$

The polymerization rate (disappearance of acrylate) is then of the form

$$R_p = k_p [R^\bullet] [M]. \quad (2.6)$$

Everything interesting is hidden in $[R^\bullet]$ and in the coefficients k_p and k_t .

2.3.1 Why k_p and k_t are not constants

As conversion rises, the mixture becomes more viscous. Small monomers may still diffuse when large macroradicals can no longer find each other by centre-of-mass motion. Termination therefore slows (*autoacceleration*: the classical Trommsdorff–Norrish gel effect), while propagation later also becomes diffusion-limited (*autodeceleration*). This picture goes back at least to analyses of diffusion-controlled termination [15] and is standard in thick-film photopolymerization models [16, 17].

At still higher conversion, radicals can “move” by adding monomer rather than by translating: *reaction-diffusion-controlled termination*, for which k_t becomes proportional to $k_p[M]$ [3, 10, 17]. Dobson and Bowman [3] emphasize a further refinement: short and long macroradicals do not share one k_t , because mobility depends on chain length and on whether the radical is already attached to the gel.

None of this is optional decoration if the process spans a wide conversion window. It *is* optional if one only asks whether a dilute flowing jet has consumed enough oxygen to gel at $\sim 2\%$ double-bond conversion [5]. The ontology must therefore allow *reduced* kinetics for a narrowly posed question, without pretending that the reduced coefficients are the true molecular rate constants of PEGDA in every geometry.

2.3.2 Thiol–ene is a different polymerization

Acrylate chain growth is not the only photocurable chemistry. Thiol–ene reactions have a step-growth character, different oxygen tolerance, and different network statistics [18]. They must not be forced into the same four-species chain-growth skeleton without rewriting both the rates and the gelation criterion. This note keeps them out of the PEGDA object.

2.4 Dose is not a state: the failure of exposure reciprocity

Photographic reciprocity says that the same $E = It$ produces the same blackening. If photopolymerization were a one-step photoreaction with no competing second-order losses, something similar might hold for p . Free-radical polymerization is not that reaction.

Bimolecular termination is second order in $[R^\bullet]$. Higher irradiance produces a higher radical concentration, hence a higher *fraction* of radicals lost to termination rather than to propagation. Oxygen inhibition is also first order in $[R^\bullet]$. The conversion therefore depends on the *shape* of $I(t)$, not only on its integral.

Wydra et al. [1] tested this on a BisGMA/TEGDMA dimethacrylate. At equal dose, different irradiances produced different conversions and shrinkage stresses; the system “does not and should not be expected” to obey a reciprocity law, because bimolecular termination dominates. Related kinetic analyses include the role of oxygen, viscosity, and time-varying intensity [2]. Dobson and Bowman [3] show a complementary symptom: the apparent scaling of R_p with I can shift from roughly $\sqrt{I}$ (classic bimolecular termination) toward a more linear I dependence when oxygen makes radical loss pseudo-first-order.

Figure 2.2 is a *cartoon* of that idea, not a plot of anyone’s FTIR data. The scientific statement is: if two illumination protocols are declared equivalent because $I_1 t_1 = I_2 t_2$, one has already assumed a constitutive law that acrylates typically violate.

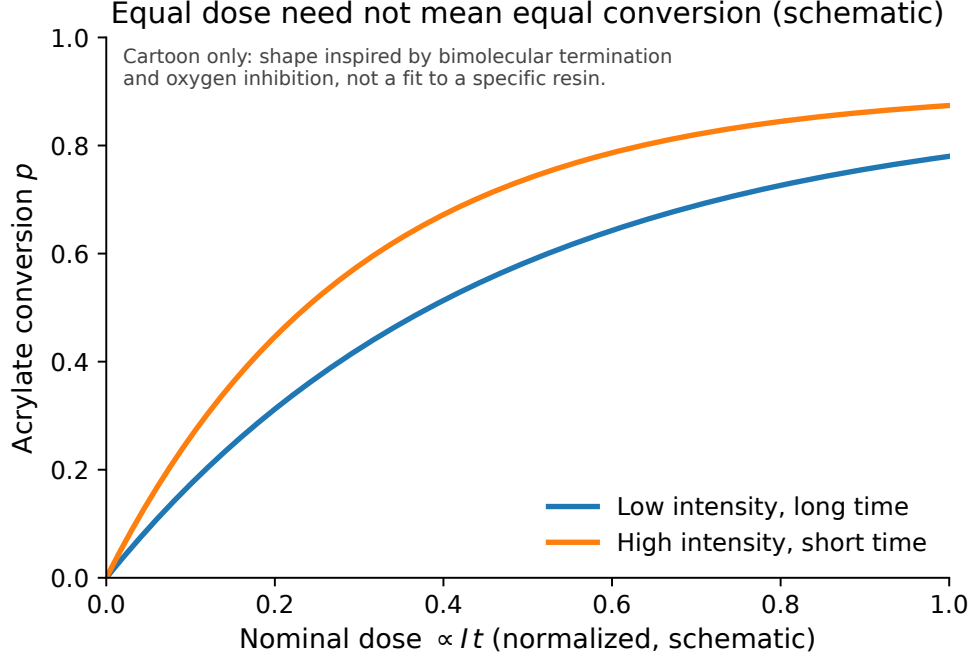


Figure 2.2: Schematic only. Two protocols with the same nominal dose need not yield the same conversion when termination is bimolecular and oxygen participates [1–3]. The curves are not experimental measurements.

2.5 Light as a field in a participating medium

Irradiance is not a property of the lamp alone. It is a field $I(\mathbf{x}, t)$ inside an absorbing (and sometimes scattering) mixture.

2.5.1 Beer–Lambert attenuation that evolves with chemistry

If scattering and emission are negligible and the beam is treated as unidirectional, a steady radiative balance reduces to

$$\frac{\partial I}{\partial z} + A(\mathbf{x}, t) I = 0, \quad (2.7)$$

with z along the beam [10]. The attenuation A is a sum of contributions: remaining photoinitiator, unreacted monomer, polymer, and any dedicated absorber dye. As initiator is consumed, A can fall (*photobleaching*, especially for Type II dyes [6]). As polymer forms, A can rise if the polymer absorbs more than the monomer [10]. In an optically thick sample the result is a polymerization *front* that advances from the illuminated face [6, 19].

Equation (2.7) is already an engineering reduction of radiative transfer. It is appropriate for a clear PEGDA resin. It is not appropriate for a ceramic-filled slurry, where scattering dominates.

2.5.2 Pixels, grayscale, and the point-spread function

A digital micromirror device (DMD) does not paint a perfectly sharp bitmap onto the resin. Each micromirror illuminates a small neighbourhood. Montgomery et al. [10] represent that

neighbourhood as a Gaussian and obtain the surface irradiance by summing all pixels, including grayscale (partial on-time of a mirror). Meenakshisundaram et al. [4] insist that the true kernel need not be Gaussian: measured irradiance from a projection head can have a flatter core and more bleed than a first-order Gaussian, because of mirror tilt, imperfect collimation, and the relay optics [citing 20].

The ontology distinction:

- the *commanded bitmap* is a discrete pattern of grayscale values;
- the *surface irradiance* is a continuous field obtained by convolution with the optical kernel and, if the head or the image moves, by a time integral;
- the *in-resin irradiance* is that surface field after attenuation (2.7);
- the *chemical conversion* is the result of the kinetic skeleton driven by the in-resin irradiance.

Collapsing all four into a Jacobs working curve throws away the objects one needs in order to predict grayscale gradients and overcure [4, 10].

2.6 Two independent motions

It is easy to say “the process is moving.” Two different velocities must be named (Fig. 2.3).

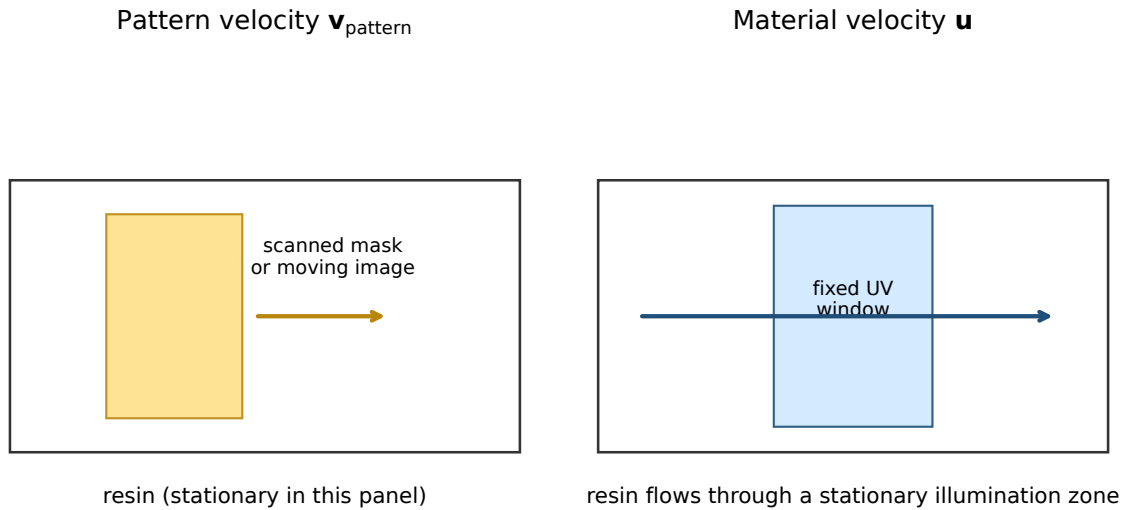


Figure 2.3: Left: the illumination pattern moves over a vat (scanning mask or scrolled image) while the resin is, to first approximation, still [4]. Right: the resin flows through a fixed light window [5]. A fluid element experiences the *relative* motion of pattern and material.

Pattern velocity $\mathbf{v}_{\text{pattern}}$. In scanning-mask projection, the projector (or the bitmap on the DMD) translates so that a large area can be addressed with a small chip [4]. Frame rate, scan speed, and projected pixel pitch together decide how long a resin point sees each column of the movie. One can even scroll the bitmap to *cancel* stage motion, so that a feature is stationary in the vat while the hardware moves.

Material velocity $\mathbf{u}$. In a microfluidic jet, oligomer is carried through a stationary UV spot of length L at speed U [5]. The residence time in the light is $t_L = L/U$. If the lamp is pulsed for a duration t_{uv} , the relevant dimensionless exposure is t_{uv}/t_L .

A material element at position $\mathbf{x}_p(t)$ with $d\mathbf{x}_p/dt = \mathbf{u}$ sees

$$I_p(t) = I(\mathbf{x}_p(t), t). \quad (2.8)$$

That sampled history is what the kinetic skeleton should be fed. Calling both “speed” hides whether one is moving the lamp, the mask, or the liquid.

2.7 Flow replenishes oxygenated resin

In a closed, stagnant film, dissolved oxygen is a finite inventory (plus whatever diffuses from the walls). In a continuous flow, fresh, oxygenated liquid continually enters the illumination zone. Slutzky et al. [5] make this the condition for fiber formation: polymerization remains negligible until oxygen is depleted, and the critical processing speed scales with how fast photolysis can eat the incoming O_2 inventory,

$$U_c \sim \frac{L \phi \varepsilon I [\text{PI}]_0}{[\text{O}_2]_0}, \quad (2.9)$$

up to the precise photophysical factors in their analysis. They also argue that, for their jet diameter and speeds, oxygen *diffusion into the jet during the residence time* is a small correction relative to advection of the dissolved inventory. That statement is geometry-specific. In a thin film against PDMS, the opposite limit can hold: diffusion from the wall dominates [13].

The ontology: *advection* and *diffusion* of oxygen are different supply routes. A model that includes only one of them is answering a different experiment.

2.8 Conversion is not gelation is not modulus

Figure 2.4 is the single most important hierarchy in this note.

2.8.1 Gelation

Gelation is the appearance of a sample-spanning covalently connected cluster: the weight-average molecular weight diverges [7, 21]. For a chain-growth polymerization of a multifunctional acrylate, that can occur at *low* conversion because each chain can connect many monomers.

Two operational definitions in the present corpus must not be mixed without comment.

- Slutzky et al. [5] take gelation, following earlier microfluidic work, when about 2% of double bonds have reacted ($[M]/[M]_0 = 0.98$). That is a convenient kinetic threshold for “the jet no longer behaves as a liquid” under their conditions.
- Zhu et al. [6] compute a conversion at gelation from a recursive Macosko–Miller argument that includes intramolecular loops. If θ is the probability that a reacted acrylate connected

Do not collapse these layers into a single “dose”

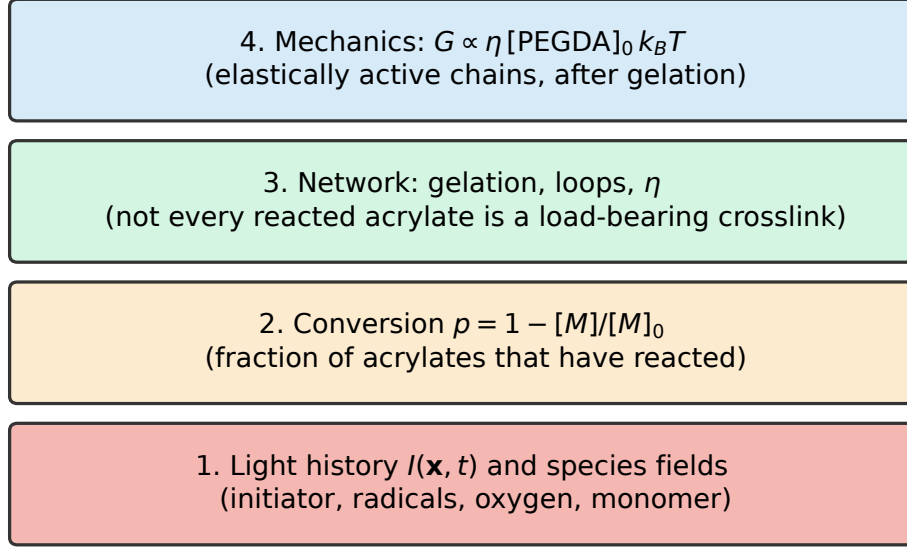


Figure 2.4: Four layers that must not be identified. Light history and species fields determine conversion p ; p and composition determine whether a sample-spanning network exists and how many chains are elastically active; modulus is a mechanical consequence of that network [6–8].

intermolecularly rather than as a loop, then

$$P_{\text{gel}} = \frac{1 - \theta}{2\theta}, \quad \theta = \frac{[\text{PEGDA}]_0}{[\text{PEGDA}]_0 + 2C_0}, \quad (2.10)$$

so $P_{\text{gel}} = C_0/[\text{PEGDA}]_0$. If the precursor is more dilute than C_0 , $P_{\text{gel}} \geq 1$: the sample never gels even at full acrylate conversion, because too many additions are wasted as loops.

Both can be legitimate. They are not the same predicate.

2.8.2 Loops and elastically active density

Even after gelation, not every reacted acrylate contributes a load-bearing chain. An intramolecular loop links a strand to itself and does not transmit stress the way an intermolecular crosslink does [6, 8]. Zhu et al. [6] construct a fractional crosslinking density η from the probabilities that arms lead to the infinite network, and then write a phantom-network shear modulus

$$G = \eta N_A [\text{PEGDA}]_0 k_B T. \quad (2.11)$$

The ideal affine or phantom network without loops is an upper bound; real PEGDA gels sit below it. Fitting G as a smooth function of dose, skipping η , hides the concentration-dependent gel point that Zhu et al. [6] used to design stiffness gradients.

2.8.3 Heat, shrinkage, and swelling

Polymerization is exothermic; thick or intensely illuminated samples heat, and rates depend on temperature [3, 16]. Conversion also shrinks the volume and, in hydrogels, the network later imbibes solvent. Those are real couplings. They are *downstream* of a trustworthy conversion field. Treating them as the first objects to predict, before $p(\mathbf{x}, t)$ is even defined, inverts the hierarchy of Fig. 2.4.

2.9 How the five campaign papers partition the ontology

The local literature folder is not five versions of one model. Each paper owns a different layer.

Table 2.1: Division of labour among the five unique reports in the campaign corpus. “Owns” means: this is the layer the paper actually closes with equations and experiments. It does not mean the other layers are false.

Report	Layer owned
Dobson and Bowman [3]	High-resolution kinetic ontology: many species, free-volume transport, short versus long radicals, oxygen, heat, one- and three-dimensional balances. Chemistry illustrated on HDDA, not PEGDA.
Montgomery et al. [10]	Spatial grayscale printing: four tracked species, conversion-dependent k_p and k_t , evolving Beer–Lambert field, pixel overlap, lateral diffusion that blurs commanded edges. PEGDA 250 / Irgacure 819.
Meenakshisundaram et al. [4]	Projection kinematics: measured (possibly non-Gaussian) pixel irradiance, scanning mask, discrete frames, dose accumulation. Cure profile compared to printed dimensions; chemistry is not the species skeleton above.
Slutzky et al. [5]	Flowing material through a fixed light window: advection of initiator, radicals, monomer, and oxygen; critical speed; pulse duration versus residence time. PEGDA 575 jet; oxygen diffusion into the jet argued small.
Zhu et al. [6]	Type II eosin Y bleaching plus conversion-to-gelation-to-modulus closure with loops. Frontal photopolymerization in depth. PEGDA 575 hydrogel; no advection.

A predictive description of a *flowing grayscale PEGDA hydrogel* needs all four chemical–physical layers—kinetics, optics, transport, network—but it should inherit each layer from the paper that actually resolved it, not average their rate constants.

2.10 What must not be identified

For a masters-level reader, the practical discipline is a list of forbidden equivalences.

1. $E = \int I dt \not\equiv$ conversion p . Reciprocity fails when termination is bimolecular and oxygen participates [1, 2].
2. Commanded bitmap $\not\equiv$ surface irradiance $\not\equiv$ in-resin $I(\mathbf{x}, t)$ [4, 10].
3. Pattern velocity $\not\equiv$ resin velocity [4, 5].

4. Dissolved oxygen inventory $\neq$ oxygen leaking through a wall [5, 13].
5. Type I photolysis $\neq$ Type II dye/amine bleaching [6, 10, 11].
6. PEGDA 250 in a dry thin film $\neq$ PEGDA 575 in water $\neq$ HDDA bulk [3, 6, 10].
7. Acrylate conversion $p \neq$ gelation $\neq$ elastically active density $\eta \neq$ modulus G [6, 8, 21].
8. A 2% double-bond threshold [5] $\neq$ the Macosko–Miller $P_{\text{gel}}([\text{PEGDA}]_0)$ [6].
9. Chain-growth acrylate kinetics $\neq$ thiol–ene step growth [18].
10. Jacobs working-curve depth [9] $\neq$ a species-resolved polymer state.

2.11 Implications for a predictive description

The ontology above already implies an order of questions, still without reference to implementation.

1. **Local constitutive response.** For a prescribed $I(t)$, how do initiator, radicals, oxygen, and acrylate evolve? This is the question Dobson and Bowman [3] answer at high chemical resolution and Montgomery et al. [10] answer with four species and conversion-dependent coefficients. Until this map is trusted—including an equal-dose test that is *allowed to fail*—spatial complexity only conceals error.
2. **Optical boundary condition.** What irradiance actually sits on $z = 0$ as a function of bitmap, pixel kernel, and pattern motion [4]? Then, how does $A(p, [\text{PI}])$ carry that field into the depth [6, 10]?
3. **Transport.** Which species move relative to the material? Oxygen almost always does; monomer may; network-bound radicals do not [3, 10]. Is the supply advective [5] or diffusive from a wall [13]?
4. **Network closure.** Given p and composition, has the sample gelled, and what is η and G [6]? This step must not be used to patch kinetic coefficients.

The campaign choice that follows from this background is conservative: treat PEGDA chain-growth as the first chemical object; treat Dobson’s full species set as a *reference* against which reduced $k_p(p)$ and $k_t(p)$ can be judged, not as the first process model; treat scanning projection and plug flow as two different kinematic experiments; treat Zhu’s loop-aware modulus as a post-processor on conversion. Those are scientific restrictions, not software preferences.

Chapter 3

Literature study and research gaps

3.1 Scope of the corpus

The working library comprises five unique journal papers, the supporting information of Dobson and Bowman [3], and a duplicate file of Meenakshisundaram et al. [4] that must not be cited twice. Supporting classics (oxygen inhibition, reciprocity, Macosko–Miller gelation, CLIP, frontal photopolymerization) are used as they appear in those reports, not as an unbounded review of photopolymer science.

The five papers do not compete to be “the PEGDA model.” Each *owns a layer* (Table 2.1 in Chapter 2). A flowing grayscale hydrogel would need all layers, inherited from the paper that actually closed each one.

3.2 High-fidelity kinetics: Dobson and Bowman

Dobson and Bowman [3] construct a first-principles description of multifunctional acrylate photopolymerization with a large species set: photoinitiator, primary radicals, short and network-bound macroradicals, unreacted and partially reacted monomer, oxygen, and terminated polymer. Mobility is tied to free volume; termination includes reaction-diffusion control and chain-length effects; heat release is optional. The illustrated chemistry is 1,6-hexanediol diacrylate (HDDA) with Irgacure 819, not PEGDA.

The campaign uses this paper as a *reference ontology*, not as production rates. The temptation to “implement Dobson first in 2-D” is the usual way a PhD project becomes un-debuggable: too many species, too many spatial operators, and no algebraic limit that can fail loudly. If, after a correct Montgomery transcription, equal-dose FTIR still cannot be organised, Dobson’s chain-length k_t is the parked refinement—not the first commit.

3.3 Production spatial kinetics: Montgomery et al.

Montgomery et al. [10] is the production constitutive law for Type I PEGDA 250 / 0.1 wt% Irgacure 819 in a thin film. Four species are tracked: initiator C_I , radicals C_R , oxygen C_O ,

monomer (acrylate groups) C_M . With diffusion,

$$\partial_t C_\alpha = \nabla \cdot (D_\alpha(p) \nabla C_\alpha) + R_\alpha, \quad (3.1)$$

and R_α given by their Eqs. 16–19. Photolysis is first order in remaining initiator, with $\beta = 2.70 \times 10^{-3} \text{ s}^2 \text{ kg}^{-1}$ multiplying I in W m^{-2} so that βI is a frequency. Two radicals are produced per cleavage ($m = 2$). Termination is $2k_t C_R^2$ plus oxygen inhibition $k_O C_O C_R$.

The closures (their Eqs. 22–27, Table 2) make k_p and k_t conversion-dependent:

$$k_p(p) = \frac{k_{p0}}{1 + (k_{p0}/k_{p,D0}) \exp(c_p p)}, \quad (3.2)$$

$$k_t(p) = \left[\frac{1}{k_{t,SD}} + \frac{\exp(c_t p)}{k_{t,TD0}} \right]^{-1} + C_{RD}(1 - p)k_p(p). \quad (3.3)$$

(The paper writes $\exp(cp)$ for $\exp(c_p p)$ with c the relative viscosity coefficient.) Harmonic liquid–solid diffusivities (Eq. 28, Table 3) drop by four orders of magnitude from liquid to solid. Beer–Lambert attenuation (Eqs. 4–6) uses molar initiator absorptivity, polymer versus monomer absorption, and an optional photoabsorber term wA_{abs} .

The optical boundary condition (Eq. 7) superposes Gaussians of radius $\sigma = 17.7 \mu\text{m}$ on a $50 \mu\text{m}$ pixel pitch, with $I_{\text{pixel}} = 41.08 \text{ W m}^{-2}$ and $I_{\text{ideal}} = 64 \text{ W m}^{-2}$. G0 is full intensity and G100 is dark. Their multilayer prints with 0.06 wt% absorber are a different optical depth than the FTIR cell.

What we did not take from this paper as gospel. The Gaussian kernel is a fallback, not a measured DMD map. The linear G0–G100 map used in code is *assumed*, not their SI RGB calibration. Table 2 rates are a fit to one thin-film recipe; they are not universal PEGDA constants. Dissolved oxygen is not a Table 2 entry; the campaign flags $C_{O,0}$ as **still required**.

3.4 Projection kinematics: Meenakshisundaram, Sturm and Williams

Meenakshisundaram et al. [4] model scanning-mask projection vat photopolymerization. The scientific object is the surface irradiance and the accumulated dose under a measured (generally non-Gaussian) pixel kernel, discrete frames, and a scan velocity. They compare a Jacobs-style cure overlay to printed dimensions; that overlay is a diagnostic, not a species conversion.

The campaign Stage 3 implements conservation and bleed tests on Montgomery’s Gaussian as a *stand-in*. A **matched** on Stage 3 requires a camera or radiometer map of the projector actually used, or digitization of their measured PSF. Scan dose scaling as $1/v$ is a dwell identity for a translating compact pattern, not a statement that p is independent of speed.

3.5 Plug flow: Slutzky, Stone and Nunes

Slutzky et al. [5] pass a sheathed aqueous jet of PEGDA 575 (54 vol%), water, dye, and 4 vol%

2-hydroxy-2-methylpropiophenone through a $L = 2.3$ mm UV slit. Steady plug flow gives

$$U \frac{dC_i}{dx} = R_i(C, k_d(x)), \quad k_d = 0 \text{ outside } [0, L]. \quad (3.4)$$

Their Eqs. 1a–d use $k_d = \phi \varepsilon I$ with I in Einstein $\text{m}^{-2} \text{s}^{-1}$, termination $k_t R^2$ (no extra factor of two), and one radical per photolysis as written in their Table 1. Taking the ratio of monomer and oxygen balances at constant k_p, k_O yields Eq. 3,

$$\frac{C_O}{C_{O,0}} = \left(\frac{C_M}{C_{M,0}} \right)^{k_O/k_p}. \quad (3.5)$$

Gelation for comparison to fiber onset is the operational predicate $M/M_0 = 0.98$ (about 2% double bonds reacted), following microfluidic convention [13]—not Zhu’s P_{gel} . Oxygen diffusion into the $40 \mu\text{m}$ jet during residence is argued small relative to advection of the dissolved inventory.

Critical speed scales as $U_c \sim L k_d [\text{PI}]_0 / [\text{O}_2]_0$. Pulsed light organises fiber length by t_{uv}/t_L with $t_L = L/U$.

This chemistry must not share a parameter book with Montgomery PEGDA 250/Irgacure.

3.6 Network mechanics: Zhu et al.

Zhu et al. [6] couple Type II eosin Y / TEOA bleaching kinetics to acrylate conversion and then to a modified Macosko–Miller recursive model with intramolecular loops. The campaign implements *only* the network post-processor (their Eqs. 21, 26, 32–38). Eosin bleaching ODEs (their Eqs. 3–8) remain off unless a Type II kinetics branch is explicitly opened.

The intermolecular probability is

$$\theta = \frac{[\text{PEGDA}]_0}{[\text{PEGDA}]_0 + 2C_0}, \quad (3.6)$$

with fitted $C_0 = 3.45$ vol% (Table A2). Then $P_{\text{gel}} = (1 - \theta)/(2\theta) = C_0/[\text{PEGDA}]_0$. If $[\text{PEGDA}]_0 \leq C_0$, the precursor never gels at $p = 1$. Without loops, $\eta = p^2$. With loops, η is assembled from probabilities that $m \geq 3$ carbon ends lead to the infinite network. The phantom modulus is $G = \eta N_A [\text{PEGDA}]_0 k_B T$, so $G/G_{\text{ideal}} = \eta$.

3.7 Supporting literature that the five papers rest on

Oxygen. Decker and Jenkins [12] established the kinetic role of O_2 as a radical scavenger. Dendukuri et al. [13] showed that PDMS walls replenish oxygen by diffusion and set a dead zone; that is a different experiment from Slutzky’s jet. Tumbleston et al. [14] turned a persistent inhibited liquid layer into CLIP.

Reciprocity. Wydra et al. [1] showed that BisGMA/TEGDMA does not obey $I \times t$ equivalence when bimolecular termination dominates. Lin et al. [2] organised related effects of oxygen,

viscosity, and time-varying intensity.

Networks. Flory [21] and Macosko and Miller [7], Miller and Macosko [22] are the gelation and post-gel recursive backbone. Zhong et al. [8] quantified how primary loops reduce elasticity. Anseth et al. [17] and Benson and North [15] frame diffusion-controlled termination. Cabral and Douglas [19] described light-driven polymerization fronts. Goodner and Bowman [16] coupled heat and mass transfer in thick films. Dadashi-Silab et al. [11] reviews Type I versus Type II initiation. Cramer and Bowman [18] marks thiol-ene as a different polymerization, parked here.

3.8 Working curves, CLIP, and why they are not the polymer state

Stereolithography practice has long used a working curve $C_d = D_p \ln(E/E_c)$ [9]. D_p is an effective optical penetration; E_c is a critical exposure at which a gel-like depth is first observed for a given resin and process. The construction is invaluable for machine calibration. It collapses oxygen, termination order, grayscale bleed, and flow into two fitted numbers. Those numbers change when the lamp spectrum, inhibitor, or speed changes, which is exactly the symptom that they are not constitutive.

Continuous liquid interface production [14] exploits a persistent oxygen-inhibited dead zone so that a part can be drawn continuously from a vat. That dead zone is a *diffusive* oxygen supply from a permeable window, closer in spirit to Dendukuri et al. [13] than to Slutzky et al. [5], whose jet carries its oxygen inventory by advection. A CLIP-style boundary condition is therefore a later Stage 4 variant, not a reason to replace p with C_d .

Frontal photopolymerization [19] is another macroscopic signature of the same microscopic rates: an optically thick, exothermic film can propagate a conversion front. Stage 4's Beer-Lambert front at $H = 200 \mu\text{m}$ is a thin, isothermal cousin of that phenomenon, not a heat-release model [16].

3.9 Undermind synthesis versus what was coded

A broader modelling note exists in the DFG Undermind workspace (unified photopolymerization roadmap and Stage 1 PEGDA implementation notes). That synthesis is right about history-dependent kinetics, independent pattern and resin velocities, and the need for a loop-aware network closure.

It is not internally consistent as an *execution order*. One passage suggests implementing Dobson first, or making the first solver two-dimensional with a measured PSF. The Stage 1 PEGDA notes, by contrast, specify a local four-species ODE, BDF integration, and Dobson as a check.

The code follows the Stage 1 notes and the local five-PDF ladder. Undermind extras (Dendukuri walls, Wydra named tests, four in-house reactors, $\sim 80 \mu\text{m}$ flowing DMD patches) are treated as

future *tests and boundary conditions*, not as extra architecture in Stages 0–6.

3.10 The gap closed by this campaign

Before this repository, the group had papers, not a citable stack. The specific gap was operational:

1. No provenance-locked parameter books, so Montgomery β could silently be used with Slutzky Einstein intensities.
2. No local ODE with named gates (dark $p = 0$, positivity, failed reciprocity, oxygen induction).
3. No constant-rate fork to test Slutzky Eq. 3 as an identity rather than a fitted curve.
4. No projection engine that is forbidden from reporting conversion.
5. No 1-D reaction–diffusion with the *same* cell RHS as the ODE.
6. No plug-flow solver on Slutzky’s stoichiometry.
7. No network post-processor that cannot retune k_p .

The remaining gap is *quantitative match* under the same formulation (Chapter 6), plus parked physics (heat, scattering, NS–gel, inverse DMD).

Chapter 4

Software architecture and numerical methods

4.1 Design principles

The package is a *staged scientific solver*, not a printer controller. Three principles are enforced in code and in process:

1. **One chemistry key per constitutive law.** Montgomery Type I PEGDA 250, Slutzky Type I PEGDA 575 jet, and Zhu Type II network parameters live in three JSON books. A driver that needs plug flow does not import Montgomery k_p .
2. **Shared RHS where the claim is the same chemistry.** Stage 4's method of lines calls the same vectorized `reaction_rates` as Stage 1. Stage 5 does *not*: Slutzky's k_d , $k_t R^2$, and Einstein I are a different stoichiometry.
3. **Post-processors cannot retune rates.** Zhu $\eta(p)$ is a map from conversion already obtained. Projection cannot emit p .

Python ≥ 3.9 is required. The installable layout is `package-dir={"src"}` in `pyproject.toml`; campaign drivers prepend `src/` to `sys.path` so a researcher can run without installing. Dependencies are NumPy, SciPy, and Matplotlib; tests use pytest.

4.2 Repository layout

The scientific kernel lives under `src/photopolymerization/`:

```
src/photopolymerization/
parameters.py          KineticParams, OpticalParams, books
kinetics/
  rhs.py               Montgomery reaction_rates, ODE rhs
  closures.py          p, kp(p), kt(p)
  protocols.py         I(t) callables
```

<code>solve.py</code>	<code>scipy.integrate.solve_ivp</code> BDF
<code>limits.py</code>	QSS residual, Slutzky Eq. 3 residual
<code>projection/psf.py</code>	Gaussian pixel kernel (no chemistry)
<code>rd/</code>	
<code>beer.py</code>	Beer-Lambert $A(p, C_I)$
<code>diffusivity.py</code>	harmonic $D_i(p)$
<code>operators.py</code>	1-D conservative diffusion
<code>mol.py</code>	method-of-lines RHS
<code>mms.py</code>	cosine manufactured solution
<code>solve.py</code>	BDF on stacked 4 n_z state
<code>flow/</code>	
<code>params.py</code>	Slutzky jet book
<code>plug.py</code>	$U \, dy/dx = R(y; k_d(x))$
<code>network/macosko.py</code>	θ, P_{gel}, η, G

Campaign artefacts sit beside the kernel, not inside it:

<code>campaigns/pegda/</code>	
<code>configs/</code>	frozen JSON per stage + three books
<code>drivers/</code>	<code>run_*.py</code> , <code>validate_*.py</code>
<code>tests/</code>	pytest gates (29 tests as of Stage 6)
<code>outputs/</code>	gitignored CSV/HDF of a run
<code>docs/figures/</code>	overview PDFs committed as lab evidence
<code>journal/entries/</code>	one markdown file per stage

4.3 The campaign quartet

Every stage $N \in \{0, \dots, 6\}$ produces the same four artefacts:

1. `config.pegda_stageN.json` — frozen knobs for *this* claim.
2. `run_pegda_stageN.py` — forward calculation and overview figure.
3. `validate_pegda_stageN.py` — named gates, OVERALL PASS or FAIL.
4. `journal/entries/2026-09-13-pegda-stageN.md` — one evidence tag.

A pytest file `test_pegda_stageN.py` wraps the same gates so a regression cannot silently reopen a closed stage. The process rule is to **stop** after a stage: engineering success (no NaN, finite p) is not a licence to couple Navier–Stokes.

4.4 Parameter books

4.4.1 Montgomery 2022 (Type I, PEGDA 250)

File `parameter_book.montgomery_2022.json`. Initiator loading 0.1 wt% Irgacure 819; $\beta = 2.70 \times 10^{-3} \text{ s}^2 \text{ kg}^{-1}$ with I in W m^{-2} ; $m = 2$. Table 2 closures k_{p0} , $k_{p,D0}$, c_p , $k_{t,SD}$, $k_{t,TD0}$, C_{RD} , k_O . Calculated $C_{M0} = 8880 \text{ mol m}^{-3}$, $C_{I0} = 2.651 \text{ mol m}^{-3}$. Dissolved oxygen $C_{O0} = 1.0 \text{ mol m}^{-3}$ is labelled **still required**: it is a plausible order of magnitude, not a FTIR-fitted inventory. Optical Table 1 coefficients and Table 3 liquid/solid diffusivities are attached for Stage 4 only.

4.4.2 Slutzky 2019 (Type I, PEGDA 575 jet)

File `parameter_book.slutzky_2019.json`. Window $L = 2.3 \text{ mm}$; Table 1 k_p , k_t , k_O ; $k_d = \phi \varepsilon I$ with I in Einstein $\text{m}^{-2} \text{ s}^{-1}$. Gel predicate $\min M/M_0 \leq 0.98$. This book must never be mixed with Montgomery βI .

4.4.3 Zhu 2020 (network only)

File `parameter_book.zhu_2020.json`. $C_0 = 3.45 \text{ vol\%}$ (Table A2, *fitted*). Flag `eosin_Y_bleaching_odes: off`. Phantom G uses an assumed PEGDA density to convert vol% to mol m^{-3} ; ratios $G/G_{\text{ideal}} = \eta$ are independent of that density.

4.5 Time integration of the local ODE

The Stage 1 state is $y = (C_I, C_R, C_O, C_M)$. The system is stiff once radicals appear: photolysis and oxygen scavenging are fast relative to acrylate conversion. The integrator is SciPy's `solve_ivp` with method BDF, relative tolerance 10^{-6} and absolute tolerances that keep C_R from being treated as zero while it is still chemically active.

A `constant_rate=True` fork freezes $k_p = k_{p0}$ and $k_t = k_t(p = 0)$. That fork exists only so Slutzky Eq. 3 and a quasi-steady radical residual can be tested as *identities*. Production runs use the full $k_p(p)$, $k_t(p)$.

4.6 Projection without chemistry

Stage 3 evaluates Montgomery Eq. 7,

$$I(x) = \sum_{\text{on}} I_{\text{pixel}} \exp\left(-\left(\frac{x - x_0}{2\sigma}\right)^2\right), \quad (4.1)$$

with $\sigma = 17.7 \mu\text{m}$, $I_{\text{pixel}} = 41.08 \text{ W m}^{-2}$, pitch $50 \mu\text{m}$. The 1-D integral of one Gaussian is $I_{\text{pixel}} 2\sigma\sqrt{\pi}$; the 2-D axisymmetric integral is $4\pi\sigma^2 I_{\text{pixel}}$ for the written kernel. Linear grayscale (G0 full, G100 dark) is an *assumption* for tests, not a camera calibration.

4.7 One-dimensional reaction–diffusion

The stacked state is length $4n_z$. Interior nodes use a conservative second-difference of $D_i(p)\partial_z C_i$. Harmonic interpolation of D_i follows Montgomery Eq. 28. Beer–Lambert attenuation along z uses

$$A = \alpha_I C_I + pA_{\text{poly}} + (1 - p)A_{\text{mon}} + wA_{\text{abs}}, \quad (4.2)$$

with w interpreted as weight percent as in their wording of Eq. 6.

An explicit Fourier time step $H^2/(2n_z^2 \max D_i)$ is *documented* as a CFL reference; the integrator remains implicit BDF. A cosine manufactured solution with frozen D_I^{liq} checks the diffusion operator independently of chemistry. Sealed Neumann boundaries conserve $\int C_M$ in the dark. An oxygen Dirichlet node at the lit face is an inhibition-layer probe, not a CLIP printer.

4.8 Plug-flow marching

Stage 5 integrates

$$U \frac{dy}{dx} = R(y; k_d(x)) \quad (4.3)$$

with $y = (\text{PI}, R, y_{\text{O}_2}, M)$ in Slutzky’s units. Outside $[0, L]$, $k_d = 0$. The independent variable is x , not t ; pulse lengths in the overview figure are kinematic Ut_{uv} . A gate that PI is frozen for $x > L$ must sample with `searchsorted`, not `argmin` of $|x - L|$ (which can still lie inside the window).

4.9 Network algebra

Stage 6 evaluates Zhu’s looking-out construction. No ODE is integrated. Montgomery $p(t)$ may be passed through $\eta(p)$ only if the plot is labelled *hypothetical*: Type I conversion is not Type II eosin Y conversion.

4.10 Two Type I right-hand sides that must not be merged

Montgomery and Slutzky both describe Type I chain growth, and both use a four-component state that looks like (C_I, C_R, C_O, C_M) . They are nevertheless different constitutive laws.

Montgomery (Stages 1 and 4). Photolysis is $dC_I/dt = -\beta IC_I$ with I in W m^{-2} and $m = 2$ radicals per cleavage. Termination is written $2k_t C_R^2$. k_p and k_t depend on p . Time is the independent variable.

Slutzky (Stage 5). Photolysis frequency is $k_d = \phi \varepsilon I$ with I in Einstein $\text{m}^{-2} \text{s}^{-1}$, and $k_d = 0$ outside the window. The paper’s Table 1 stoichiometry is used as written: one radical per photolysis in the radical balance, termination $k_t R^2$ without an extra factor of two. Rates are constant (no $k_p(p)$). The independent variable is x at fixed U .

A shared function name `rhs` would have been a silent recipe mix. The package therefore keeps `kinetics/rhs.py` and `flow/plugin.py` as separate modules and refuses to load Montgomery β into the jet book.

4.11 What is deliberately absent

There is no Navier–Stokes solver, no radiative transfer with scattering, no inverse grayscale optimizer, no Dobson free-volume species set in production, and no mixing of eosin bleaching into `reaction_rates`. Those absences are scope locks, not unfinished TODOs of Stages 0–6.

Chapter 5

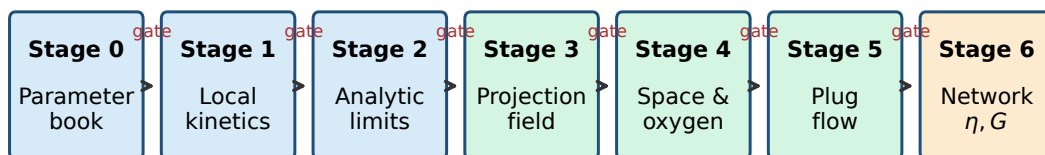
Results of the staged campaign

This chapter reports what the solvers actually did. Every numerical value is tagged by the stage that produced it. Internal gates that passed are **validated**. No row in Table 5.1 is **matched**.

Table 5.1: Campaign status on 13 September 2026. Internal **validated** is not a literature overlay.

Stage	Tag	Claim in one sentence
0	validated	Every Montgomery coefficient has a source or a still required flag.
1	validated	Local Type I ODE: dark $p = 0$, failed reciprocity, oxygen induction.
2	validated	Constant-rate fork satisfies Slutzky Eq. 3 and QSS radicals.
3	validated	Gaussian pixels conserve analytic integrals; no chemistry.
4	validated	1-D RD + Beer–Lambert: front, dead zone, PSF-dominated strip.
5	validated	Slutzky plug flow: Fig. 3 $M/M_0 \simeq 0.86$; U_c scales with k_d .
6	validated	Zhu loops: $P_{\text{gel}} = C_0/c$; $\eta(p = 1) = 0.378$ at 20 vol%.

Campaign ladder (stop at each gate; do not skip)



Parked after Stage 6: moving pattern, free-volume Dobson chemistry, heat, inverse design.

Figure 5.1: Locked stage ladder. Parked items (Dobson production RHS, NS–gel, inverse DMD, Type II bleaching, Jacobs as polymer state) remain parked.

5.1 Stage 0 — provenance before kinetics

The Montgomery book contains twenty entries with a chemistry key that forbids mixing with Zhu eosin Y or Slutzky Einstein intensities. β is stored as $\text{s}^2 \text{kg}^{-1}$ so that βI is a frequency when I is in W m^{-2} ; using milliwatts per square centimetre without conversion would shift photolysis by a factor of ten. $C_{O0} = 1.0 \text{ mol m}^{-3}$ is explicit and `still required`.

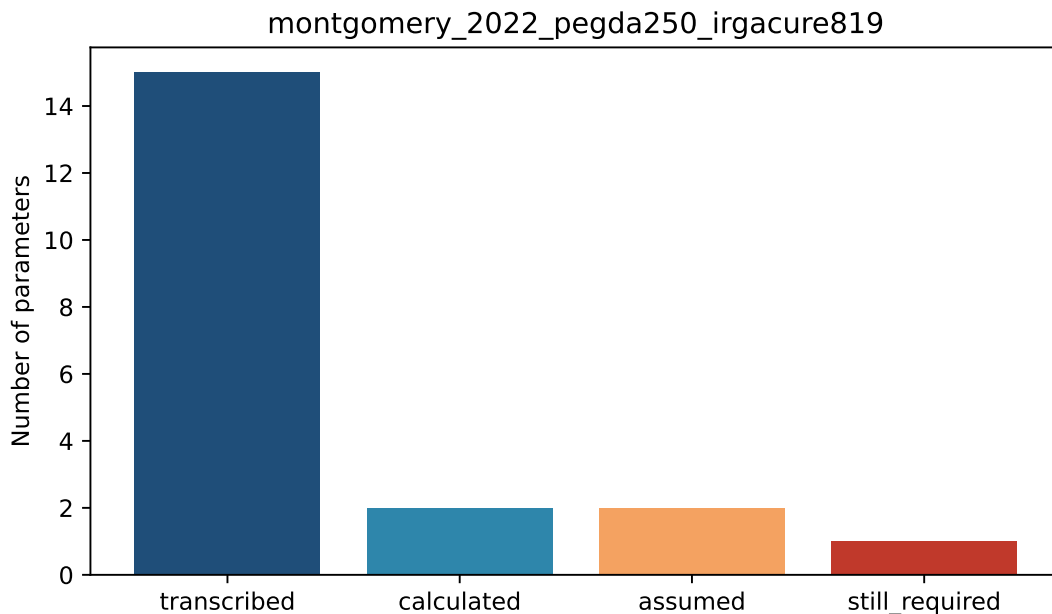


Figure 5.2: Stage 0 provenance overview (teaching layout of the book, not an experiment).

5.2 Stage 1 — local Montgomery ODE

With $I = 0$, conversion remains $p = 0$. Species stay non-negative to a BDF floor of order $10^{-14} \text{ mol m}^{-3}$. The acrylate integral residual against $\int R_p dt$ is $\sim 5 \times 10^{-5}$. Equal dose does not give equal conversion: 64 W m^{-2} for 20 s yields $p = 0.740$, while 16 W m^{-2} for 80 s yields $p = 0.800$. Oxygen delays the $p = 0.02$ crossing from 0.90 s to 2.11 s under the assumed C_{O0} .

These numbers organise the *sign* of reciprocity failure [1] and oxygen induction. They are not a digitization of Montgomery’s FTIR figure: intensity, film thickness, and dissolved oxygen of that figure were not re-fitted here.

5.3 Stage 2 — analytic identities on a constant-rate fork

While $C_O/C_{O0} > 0.05$, Slutzky Eq. 3 holds to a relative residual 1.12×10^{-6} on the frozen- k_p fork. The same residual on the full Montgomery closures is 1.10×10^{-6} in that window because $k_O/k_p \sim 1.9 \times 10^3$ and conversion is still near zero. The fork is chemically real later: $k_p(0.8)/k_p(0) = 7 \times 10^{-4}$. With $C_{O0} = 0$, the quasi-steady radical residual against $\sqrt{R_i/2k_t}$ is 2.1×10^{-3} after 0.5 s.

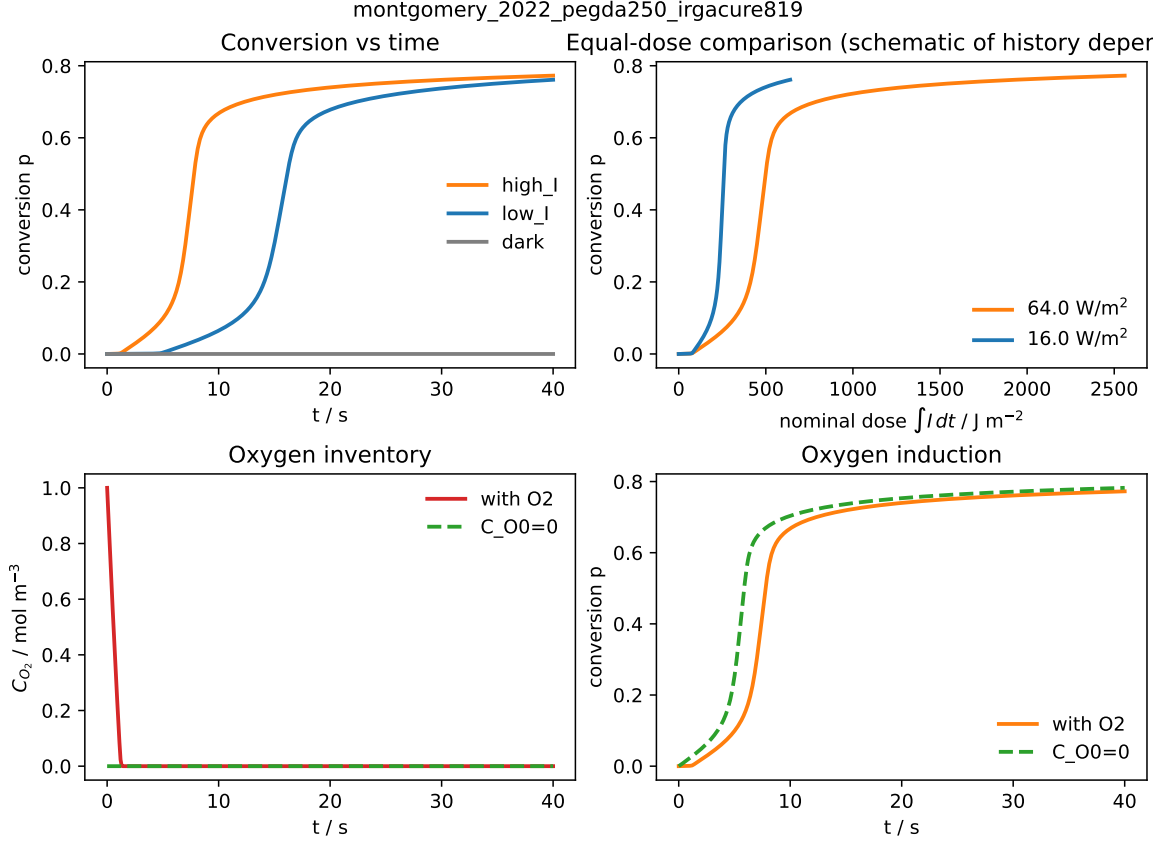


Figure 5.3: Stage 1 local ODE: conversion histories, equal-dose pair, and oxygen induction.

Stage 2 is an identity test of the transcription. It stays **validated** by construction; there is no laboratory overlay that would make it **matched** without confusing algebra with data.

5.4 Stage 3 — Gaussian projection

On a 1201-point line, the one-dimensional pixel integral matches $I_{\text{pixel}} 2\sigma\sqrt{\pi}$ to machine precision. The two-dimensional integral matches $I_{\text{pixel}} 4\pi\sigma^2$ to $\sim 10^{-16}$. A dark bitmap is identically dark. A ten-pixel block bleeds: edge 25.4 W m^{-2} , interior 50.8 W m^{-2} . Scan dose peaks scale as $1/v$ to 2×10^{-2} relative.

The integral of I is *not* I_{pixel} times the geometric pixel area $(50 \mu\text{m})^2$, because the written kernel is a Gaussian of width σ , not a top-hat of the pitch. G0/G100 linearity is assumed. A measured PSF [4] is still required for **matched**.

5.5 Stage 4 — reaction–diffusion and Beer–Lambert

A surface cell with $D_i = 0$ reproduces Stage 1 conversion to $|\Delta p| \sim 4 \times 10^{-6}$ at 8 s and $I = 64 \text{ W m}^{-2}$, so the MOL chemistry is the ODE chemistry. Dark sealed films conserve $\int C_M$ to a relative residual of zero. A cosine manufactured solution with frozen liquid D_I has relative max error 6.9×10^{-3} .

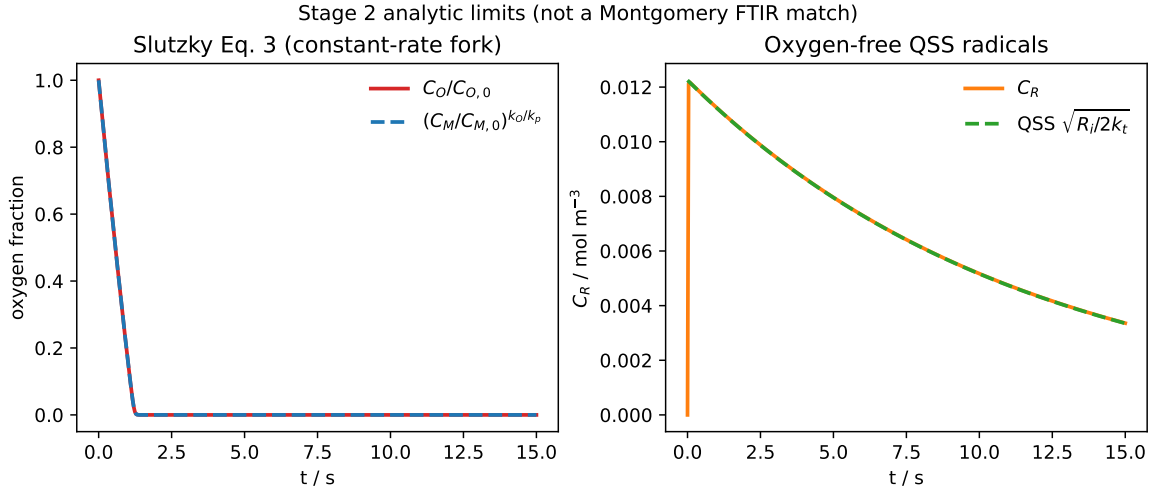


Figure 5.4: Stage 2: oxygen–monomer identity and quasi-steady radical residual.

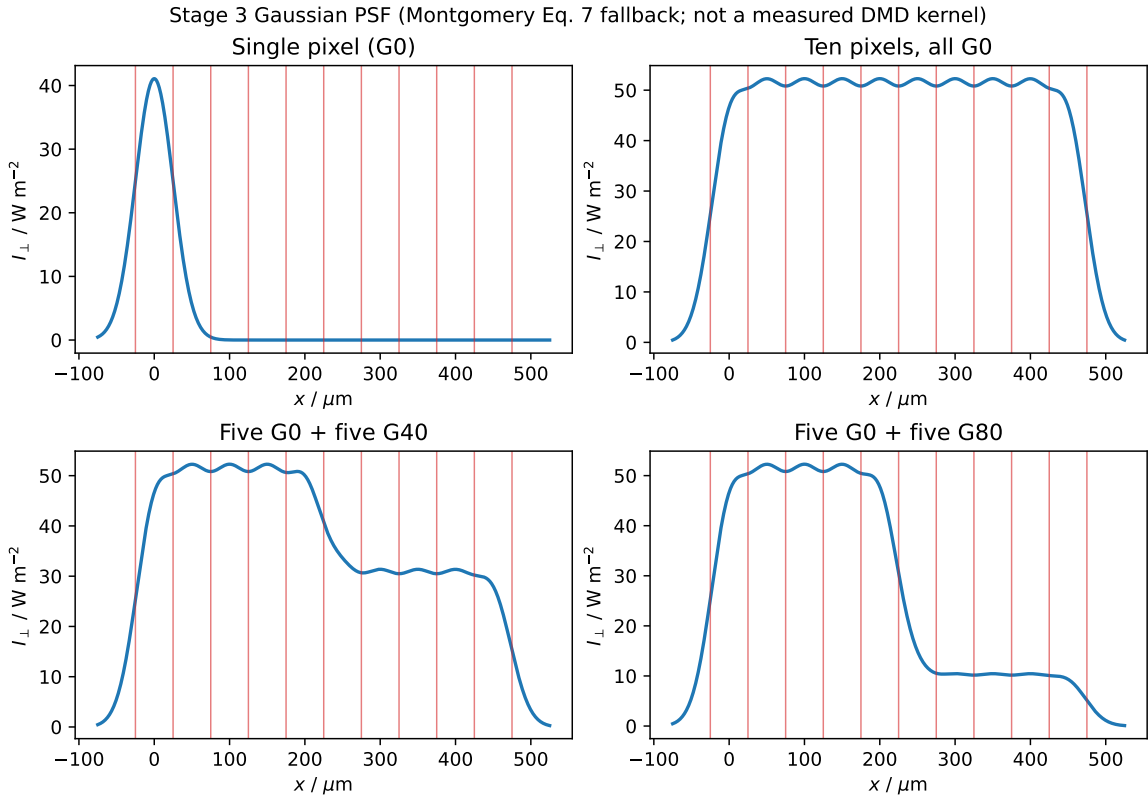


Figure 5.5: Stage 3 Gaussian projection (Montgomery Eq. 7 fallback). No conversion is plotted.

A labelled $H = 200\ \mu\text{m}$ film with $w = 0.06\ \text{wt}\%$ absorber reaches optical thickness $\beta_{\text{opt}} = 1.50$ and $p(0) - p(H) = 0.52$ at 8 s. That geometry is *not* Montgomery’s FTIR cell. An oxygen Dirichlet condition at the lit face gives $p_{\text{wall}} = 0.0069$ versus $p_{\text{mid}} = 0.53$.

A G0|G80 conversion strip has 10–90% width $53\ \mu\text{m}$, larger than $\sigma = 17.7\ \mu\text{m}$. Turning Table 3 diffusivities on versus off changes that width by only $0.21\ \mu\text{m}$ at 8 s. The 2–3 pixel printed interface in Montgomery Fig. 7 is therefore consistent with a *PSF-dominated* blur, not with radical diffusion as the leading length. A campaign gate that demanded “ $D_{\text{on}} \gg D_{\text{off}}$ ” would have been scientifically false; it was replaced by “width $> \sigma$ ” plus a note.

There is no coupled x – z print grid in this stage.

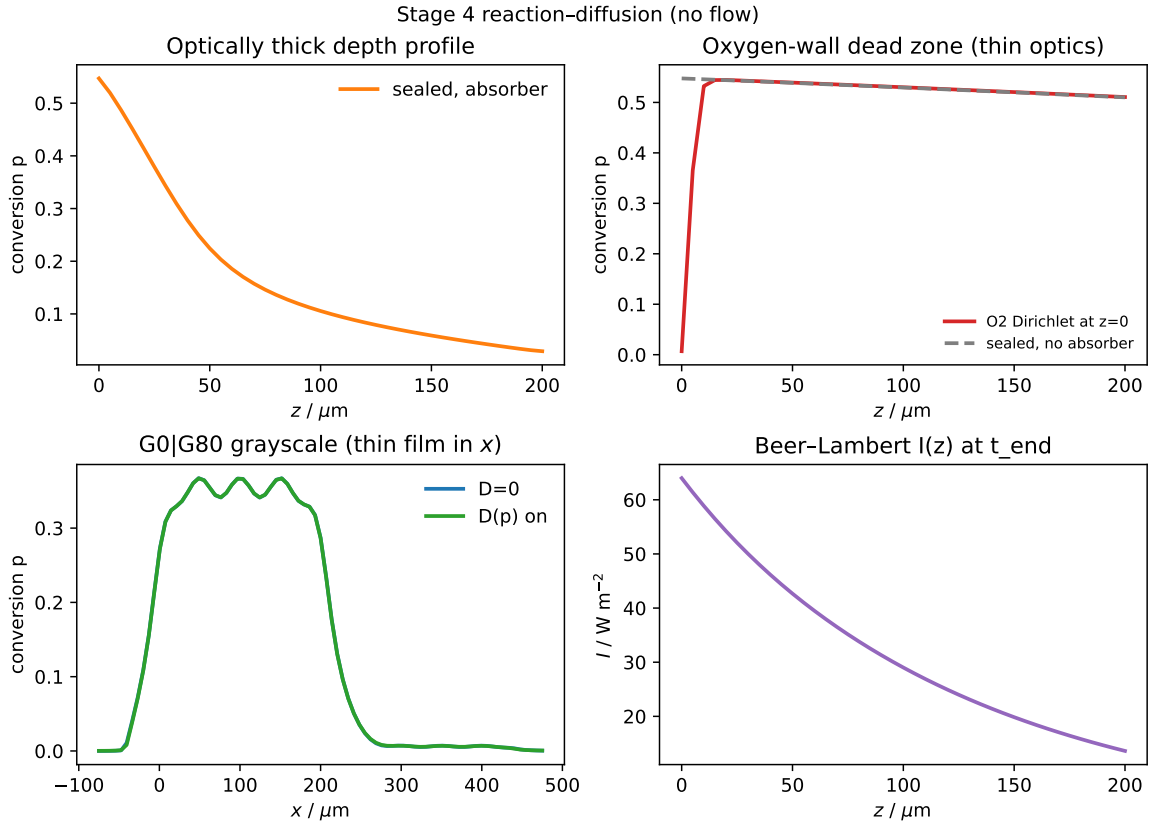


Figure 5.6: Stage 4: optical front, oxygen wall, and G0|G80 strip width.

5.6 Stage 5 — Slutsky plug flow

High-oxygen Fig. 2 parameters at $U = 0.003\ \text{m s}^{-1}$ keep $\min M/M_0 = 1.00$ (no gel under the 2% predicate). Low-oxygen Fig. 3 parameters reach $\min M/M_0 = 0.863$, against the paper’s quoted $\simeq 0.86$. Eq. 3 along x has residual 7.9×10^{-7} . At $U = 1\ \text{m s}^{-1}$ the jet exits unconverted. Quadrupling k_d raises U_c from 0.0018 to $0.0072\ \text{m s}^{-1}$; doubling $[\text{PI}]_0$ raises U_c from 0.0023 to $0.0045\ \text{m s}^{-1}$.

Pulse “fiber” lengths in the overview figure are Ut_{uv} , a kinematic construction, not a measured fiber campaign. Wall oxygen diffusion is off, following their $\sqrt{Dt} \ll w_1$ argument. The gel predicate is Slutsky’s $M/M_0 = 0.98$, not Zhu P_{gel} .

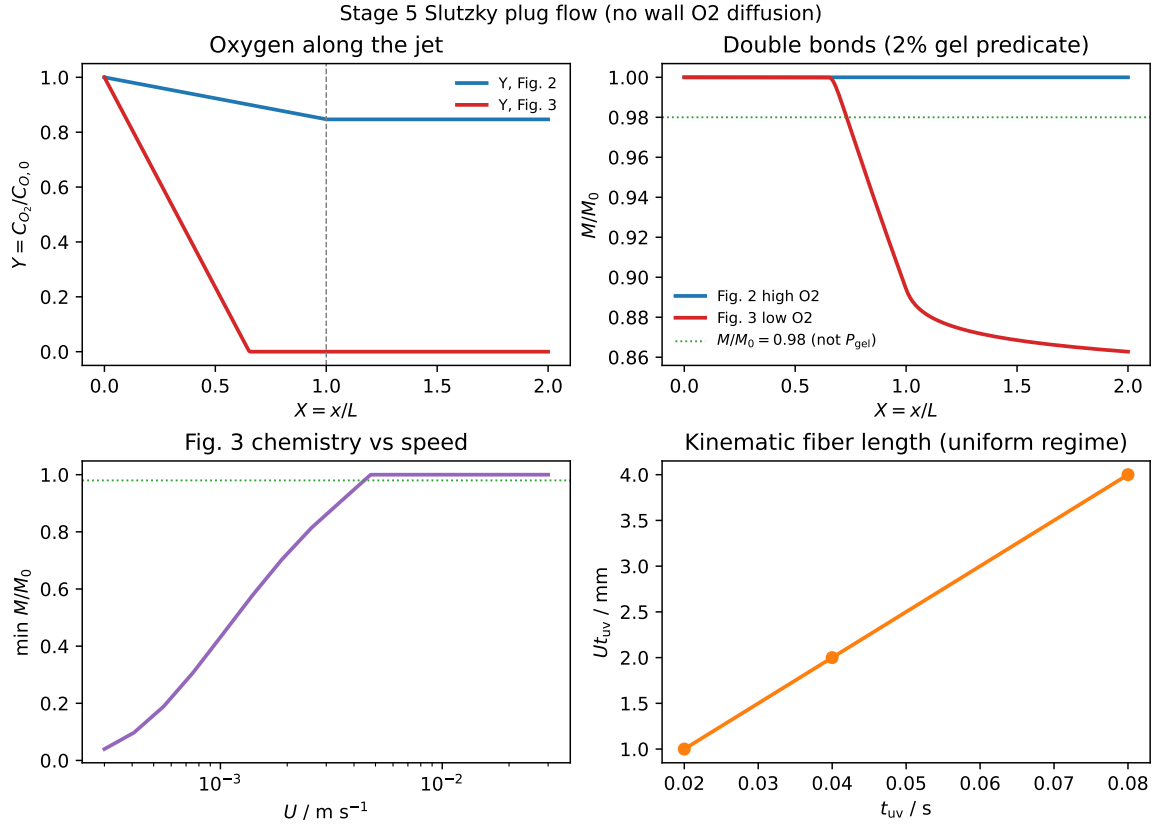


Figure 5.7: Stage 5 plug flow on the Slutzky book: conversion along x , U_c trends, pulses.

5.7 Stage 6 — Zhu network post-processor

At 20 vol% PEGDA, $P_{\text{gel}} = C_0/c = 0.1725$. If $c \leq C_0$, $\eta(p=1) = 0$: the precursor cannot gel. The no-loop identity $\eta = p^2$ holds. With loops at 20 vol% and $p = 1$, $\eta = 0.378 = G/G_{\text{ideal}}$. Pregel $\eta = 0$ for $p < P_{\text{gel}}$. $P_{\text{gel}} = 0.173$ is not Slutzky's 2% double-bond operational gel.

Passing a Montgomery $p(t)$ through η is labelled hypothetical: Irgacure Type I conversion is not eosin Y Type II conversion.

5.8 Cross-stage reasonability

Taken together, the stack behaves like photopolymerization rather than like a random ODE:

- Dark resin does not convert.
- Oxygen delays conversion; fast flow can prevent gel under an operational 2% threshold.
- Equal dose is not equal p once k_t is bimolecular and conversion-dependent.
- Pixels bleed; scan dose falls as $1/v$.
- Optically thick films convert more at the lit face; a Dirichlet O₂ wall stays almost unconverted.
- Dilute PEGDA need not gel at $p = 1$ if loops consume junctions.

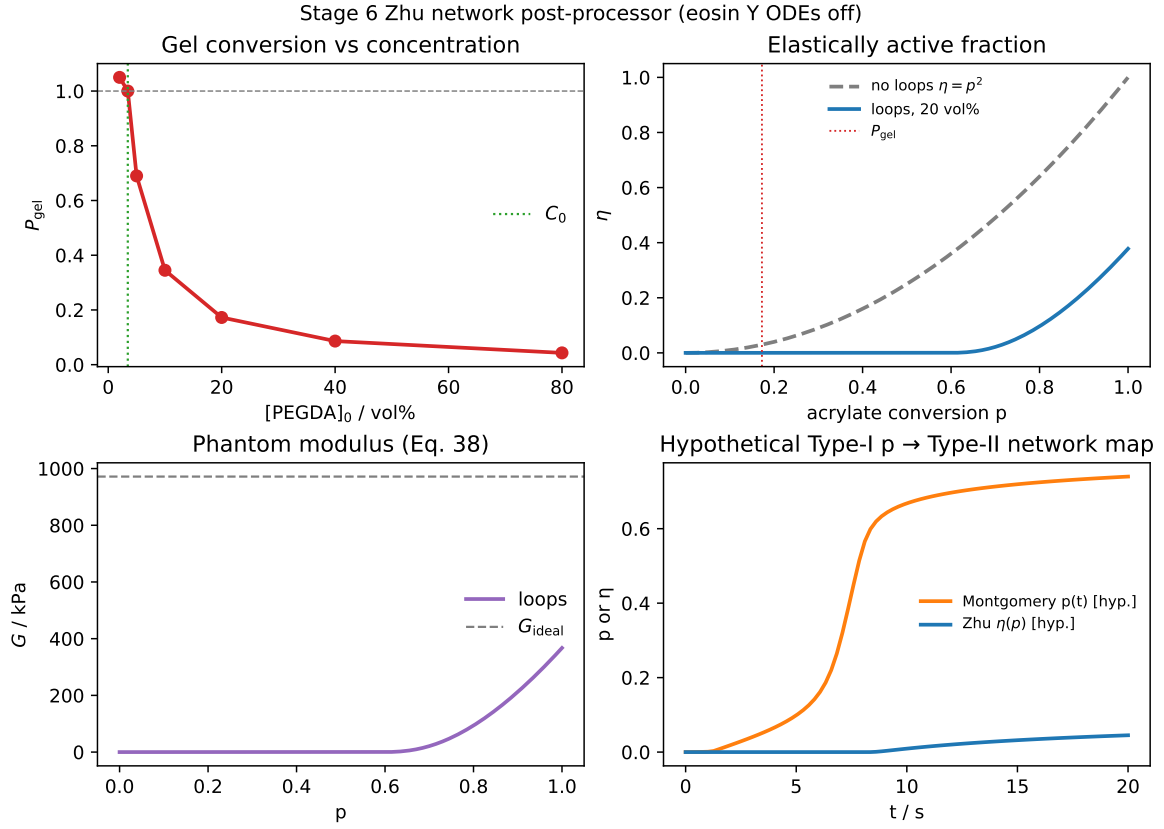


Figure 5.8: Stage 6 Macosko–Miller loops: $P_{\text{gel}}(c)$, $\eta(p)$, and a hypothetical Montgomery trajectory.

None of those statements licenses transferring a fitted number across books.

5.9 Tests

As of Stage 6, `python3 -m pytest campaigns/pegda/tests -q` reports 29 passed. Validators print **OVERALL PASS** on both `--smoke` and full modes for Stages 0–6. Outputs under `outputs/pegda_stageN_out/` are gitignored; figures under `docs/figures/` are the citable lab plots.

Chapter 6

What would constitute a literature match

A stage may be retagged **matched** only when a *named number* from a *named figure or table*, under the *same formulation*, lies inside a *pre-declared band*, and the validator plus journal are updated in the same commit of scientific record.

Matching is not “the curve looks like polymerization.” It is not transferring Montgomery k_p onto a Slutzky jet. It is not calling $M/M_0 = 0.98$ a Flory gel point.

6.1 Recommended order

1. Stage 1 FTIR overlay (Montgomery PEGDA 250 / Irgacure 819).
2. Stage 3 measured PSF (this projector, or Meenakshisundaram’s map if claiming that printer).
3. Stages 5 and 6 in parallel (different chemistries; no shared k_p).
4. Stage 4 last, after the PSF is honest: otherwise a wrong kernel will be absorbed into D_R .

Stage 2 remains **validated**: it is an identity, not a laboratory object.

6.2 Stage 1

Digitize Montgomery FTIR $p(t)$ at their reported I and film. Declare a band (for example $\max |p_{\text{sim}} - p_{\text{FTIR}}| < 0.05$ on the illuminated window, or a tighter band on the induction time). Measure or independently estimate C_{O0} ; do not hide oxygen as a free fit that also absorbs errors in β . Run the equal-dose pair at *their* two intensities, not only at 64 and 16 W m⁻².

In-house AVT resin is a *new book*, not a Montgomery match.

6.3 Stage 3

Replace Eq. 7 with a camera or radiometer single-pixel map, or digitize the measured kernel in Meenakshisundaram et al. [4] if that is the claimed device. If grayscale G0–G100 is claimed against Montgomery Fig. 3, transcribe their SI $\text{RGB} \rightarrow I$ calibration; do not keep the linear placeholder.

6.4 Stage 4

Only after Stage 3. Compare a print geometry they actually used (layer height, absorber loading, grayscale pair) to Fig. 7 interface width. Do not inflate D_R to manufacture a 100–150 μm chemical blur that Table 3 plus an 8 s strip does not produce.

Dendukuri PDMS-wall dead zones are a different experiment (permeable walls, different resin). They may become a later boundary-condition test, not a Montgomery match.

6.5 Stage 5

Overlay experimental $U_c(I, [\text{PI}]_0)$ and fiber length versus t_{uv}/t_L from Slutzky et al. [5] on the *Slutzky* book. The numerical Fig. 3 $M/M_0 \simeq 0.86$ is encouraging but is still an ODE illustration until the critical-speed map is overlaid with a declared band.

6.6 Stage 6

Overlay $G(p)$ or G/G_{ideal} versus $[\text{PEGDA}]_0$ from Zhu et al. [6] on the eosin Y aqueous gel. Do not call a hypothetical Montgomery $p \rightarrow \eta$ curve a Zhu match.

6.7 Forbidden upgrades

- Matching Stage 1 never upgrades Stage 6.
- Matching Stage 5 never upgrades Stage 1.
- A Jacobs working curve overlay never upgrades conversion.
- A pretty $G(t)$ from mixing Type II bleaching into Type I rates is *invalid*.

Chapter 7

Conclusions, limitations, and parked work

7.1 What this campaign established

A citable Python stack now exists in which PEGDA Type I conversion, Gaussian projection, one-dimensional reaction–diffusion, plug flow, and loop-aware network mechanics are *separate claims* with frozen books and named gates. Stages 0–6 are internally **validated**. The chemical ontology of Chapter 2 is enforced in code: dose is not conversion; conversion is not gelation; gelation is not modulus; Type I is not Type II; Montgomery PEGDA 250 is not Slutzky PEGDA 575; Slutzky’s 2% operational gel is not Zhu P_{gel} .

The most important *scientific* (not merely numerical) findings among the internal gates are:

1. Equal-dose reciprocity already fails in the Montgomery closures at the two intensities tested.
2. Slutzky Eq. 3 is an identity of constant k_O/k_p during induction, recovered both in time (Stage 2) and along a jet (Stage 5).
3. At Table 3 diffusivities and 8 s, radical diffusion does not set grayscale feature size; the Gaussian PSF does.
4. A 20 vol% PEGDA network with Zhu’s fitted C_0 reaches only $\eta = 0.378$ at full conversion; below C_0 it never gels.

7.2 Limitations

No FTIR, camera PSF, critical-speed map, or rheology overlay has been performed. C_{O0} remains assumed. The Gaussian kernel and linear grayscale are placeholders. Stage 4 is one-dimensional. Stage 5 pulse lengths are kinematic. Stage 6 G uses an assumed density for absolute pascals; only G/G_{ideal} is density-free. The Montgomery $p \rightarrow \eta$ plot is a thought experiment.

The literature study is bounded by the campaign PDFs plus the classics those papers cite. It is not a complete review of stereolithography, CLIP, two-photon lithography, or thiol–ene

materials.

7.3 Parked work (do not auto-start)

Full Dobson species as a production RHS; Navier–Stokes coupled to gelation; scattering; inverse DMD design; PNIPAM; thiol–ene; eosin Y bleaching ODEs mixed into Irgacure Type I; Jacobs depth as polymer state; four in-house reactors as extra architecture.

7.4 Immediate scientific next step

The first match attempt should be Stage 1: an FTIR $p(t)$ CSV on a named PEGDA / Type I recipe, a declared error band, and a validator flag `--match` that cannot pass if the book chemistry key disagrees. Until that exists, cite this report as a *modelling and verification* thesis, not as a materials-property paper.

Chapter 8

Nomenclature

Symbols are SI unless a paper uses Einstein units for photon flux; those cases are flagged in the Slutzky book.

Table 8.1: Principal symbols.

Symbol	Unit	Meaning
I	W m^{-2} or Einstein $\text{m}^{-2} \text{s}^{-1}$	Irradiance or molar photon flux
E	J m^{-2}	Dose $\int I dt$; not a material state
p	—	Acrylate conversion $1 - C_M/C_{M0}$
C_I, C_R, C_O, C_M	mol m^{-3}	Initiator, radicals, oxygen, acrylate groups
β	$\text{s}^2 \text{kg}^{-1}$	Montgomery photolysis coefficient
m	—	Radicals per photolysis ($m = 2$ Montgomery)
k_p, k_t, k_O	various	Propagation, termination, oxygen scavenging
A	m^{-1}	Beer–Lambert attenuation
σ	m	Gaussian pixel radius (Montgomery Eq. 7)
$D_i(p)$	$\text{m}^2 \text{s}^{-1}$	Harmonic liquid–solid diffusivity
U	m s^{-1}	Plug speed
L	m	UV window length
k_d	s^{-1}	Slutzky photolysis frequency $\phi \varepsilon I$
U_c	m s^{-1}	Critical speed for operational gel
θ	—	Intermolecular linking probability
C_0	$\text{vol}\%$	Zhu loop parameter (fitted 3.45 vol%)
P_{gel}	—	$C_0/[\text{PEGDA}]_0$
η	—	Elastically active density / G_{ideal}
G	Pa	Phantom shear modulus
$\mathbf{v}_{\text{pattern}}$	m s^{-1}	Scan or mask velocity
$\mathbf{u}$	m s^{-1}	Resin velocity

Chapter 9

Appendix: verification log and file inventory

9.1 How to reproduce the internal gates

From the repository root, with Python 3.9+ and the dependencies in `pyproject.toml`:

```
export PYTHONPATH=src:campaigns/pegda
python3 campaigns/pegda/drivers/validate_pegda_stage0.py
python3 campaigns/pegda/drivers/validate_pegda_stage1.py
python3 campaigns/pegda/drivers/validate_pegda_stage2.py
python3 campaigns/pegda/drivers/validate_pegda_stage3.py
python3 campaigns/pegda/drivers/validate_pegda_stage4.py
python3 campaigns/pegda/drivers/validate_pegda_stage5.py
python3 campaigns/pegda/drivers/validate_pegda_stage6.py
python3 -m pytest campaigns/pegda/tests -q
```

Each validator prints per-gate lines and **OVERALL PASS** or **FAIL**. Smoke mode `--smoke` is a shorter integration window used in continuous checks.

9.2 Overview figures

Committed teaching and gate figures (all under `docs/figures/`):

- background species, reciprocity, two-velocity, and network-stack cartoons;
- the stage-ladder diagram;
- `pegda_stage0_overview.pdf` through `pegda_stage6_overview.pdf`.

9.3 Journal entries

One markdown file per stage, all dated 13 September 2026, under `journal/entries/` with slugs `pegda-stage0` through `pegda-stage6`. This thesis does not replace those entries; it synthesises them.

9.4 What is gitignored

Publisher PDFs in `Literature/` and numerical dumps in `outputs/` are local. Do not redistribute publisher PDFs with this report.

Bibliography

- [1] James W. Wydra, Neil B. Cramer, Jeffrey W. Stansbury, and Christopher N. Bowman. The reciprocity law concerning light dose–relationships applied to BisGMA/TEGDMA photopolymers: theoretical analysis and experimental characterization. *Dental Materials*, 30(6):605–612, 2014. doi: 10.1016/j.dental.2014.02.021.
- [2] Jui-Teng Lin, Hsia-Wei Liu, Kuen-Tse Chen, and Da-Chuan Cheng. Modeling the kinetics, curing depth, and efficacy of radical-mediated photopolymerization: the role of oxygen inhibition, viscosity, and dynamic light intensity. *Frontiers in Chemistry*, 7:760, 2019. doi: 10.3389/fchem.2019.00760.
- [3] Adam L. Dobson and Christopher N. Bowman. A comprehensive, multidimensional first-principles model for free-radical photopolymerizations in bulk and thin films. *Advanced Functional Materials*, 34(22):2312607, 2024. doi: 10.1002/adfm.202312607.
- [4] Viswanath Meenakshisundaram, Logan D. Sturm, and Christopher B. Williams. Modeling a scanning-mask projection vat photopolymerization system for multiscale additive manufacturing. *Journal of Materials Processing Technology*, 279:116546, 2020. doi: 10.1016/j.jmatprotec.2019.116546.
- [5] Malcolm Slutzky, Howard A. Stone, and Janine K. Nunes. A quantitative study of the effect of flow on the photopolymerization of fibers. *Soft Matter*, 15(46):9553–9564, 2019. doi: 10.1039/c9sm01485c.
- [6] Hongyuan Zhu, Xiaoxiao Yang, Guy M. Genin, Tian Jian Lu, Feng Xu, and Min Lin. Modeling the mechanics, kinetics, and network evolution of photopolymerized hydrogels. *Journal of the Mechanics and Physics of Solids*, 142:104041, 2020. doi: 10.1016/j.jmps.2020.104041.
- [7] Christopher W. Macosko and Douglas R. Miller. A new derivation of average molecular weights of nonlinear polymers. *Macromolecules*, 9(2):199–206, 1976. doi: 10.1021/ma60050a003.
- [8] Mingjiang Zhong, Rui Wang, Ken Kawamoto, Bradley D. Olsen, and Jeremiah A. Johnson. Quantifying the impact of molecular defects on polymer network elasticity. *Science*, 353(6305):1264–1268, 2016. doi: 10.1126/science.aag0184.
- [9] Paul F. Jacobs. *Rapid Prototyping & Manufacturing: Fundamentals of Stereolithography*. Society of Manufacturing Engineers, Dearborn, MI, 1992.

- [10] S. Macrae Montgomery, Craig M. Hamel, Jacob Skovran, and H. Jerry Qi. A reaction–diffusion model for grayscale digital light processing 3D printing. *Extreme Mechanics Letters*, 53:101714, 2022. doi: 10.1016/j.eml.2022.101714.
- [11] Sajjad Dadashi-Silab, Sean Doran, and Yusuf Yagci. Photoinitiated radical polymerization through the use of photoinitiators. *Chemical Reviews*, 116(17):10212–10275, 2016. doi: 10.1021/acs.chemrev.5b00586.
- [12] Christian Decker and Andrew D. Jenkins. Kinetic approach of oxygen inhibition in ultraviolet- and laser-induced polymerizations. *Macromolecules*, 18(6):1241–1244, 1985. doi: 10.1021/ma00148a034.
- [13] Dhananjay Dendukuri, Priyadarshi Panda, Ramin Haghgooie, Ju Min Kim, T. Alan Hatton, and Patrick S. Doyle. Modeling of oxygen-inhibited free radical photopolymerization in a PDMS microfluidic device. *Macromolecules*, 41(22):8547–8556, 2008. doi: 10.1021/ma801219w.
- [14] John R. Tumbleston, David Shirvanyants, Nikita Ermoshkin, Rima Janusziewicz, Ashley R. Johnson, David Kelly, Kai Chen, Robert Pinschmidt, J. P. Rolland, Alexander Ermoshkin, Edward T. Samulski, and Joseph M. DeSimone. Continuous liquid interface production of 3D objects. *Science*, 347(6228):1349–1352, 2015. doi: 10.1126/science.aaa2397.
- [15] Sidney W. Benson and A. M. North. The kinetics of free radical polymerization under conditions of diffusion-controlled termination. *Journal of the American Chemical Society*, 81(6):1339–1345, 1959. doi: 10.1021/ja01515a013.
- [16] Michael D. Goodner and Christopher N. Bowman. Development of a comprehensive free radical photopolymerization model incorporating heat and mass transfer effects in thick films. *Chemical Engineering Science*, 57(5):887–900, 2002. doi: 10.1016/S0009-2509(01)00287-1.
- [17] Kristi S. Anseth, Christopher M. Wang, and Christopher N. Bowman. Kinetic evidence of reaction diffusion during the polymerization of multi(meth)acrylate monomers. *Macromolecules*, 27(3):650–655, 1994. doi: 10.1021/ma00081a004.
- [18] Neil B. Cramer and Christopher N. Bowman. Kinetics of thiol–ene and thiol–acrylate photopolymerizations with real-time Fourier transform infrared. *Journal of Polymer Science Part A: Polymer Chemistry*, 39(19):3311–3319, 2001. doi: 10.1002/pola.1314.
- [19] João T. Cabral and Jack F. Douglas. Propagating waves of network formation induced by light. *Polymer*, 46(12):4230–4241, 2005. doi: 10.1016/j.polymer.2005.02.051.
- [20] Chi Zhou and Yong Chen. Calibrating large-area mask projection stereolithography for its accuracy and resolution improvements. *Proceedings of the Solid Freeform Fabrication Symposium*, pages 82–95, 2009. Discussed in Meenakshisundaram et al. as evidence that DMD irradiance is not purely Gaussian.
- [21] Paul J. Flory. Molecular size distribution in three dimensional polymers. I. Gelation. *Journal of the American Chemical Society*, 63(11):3083–3090, 1941. doi: 10.1021/ja01856a061.

- [22] Douglas R. Miller and Christopher W. Macosko. A new derivation of post gel properties of network polymers. *Macromolecules*, 9(2):206–211, 1976. doi: 10.1021/ma60050a004.